Vetting the 4.8-day Transit Candidate on the Nearby M Dwarf L 198-93 (TIC 264151965)

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Abstract

Murray (2026a) posted a 4.8158 d transit candidate on L 198-93 (TIC 264151965), an M dwarf at 21.2 pc, and pre-registered a test with TESS Sectors 113–114. My search of 3,883 nearby red dwarfs recovered it; here I vet it. Six TESS sectors (2021–2026) show 29 transits with depth 1,255 ± 66 ppm, duration 1.16 h and fitted radius 1.16 (+0.05/–0.04) R⊕. Odd and even depths agree, there is no secondary eclipse, and the dip is in the raw flux. Pixel fits in all six sectors place it on the target and rule out every Gaia star within 150″ brighter than G = 20.5, including the star 106″ away to which Eumemic (2026) attributed it. TRICERATOPS gives FPP = 0.079 ± 0.004, or 0.020 ± 0.009 with MOLUSC companion limits, above the 0.015 validation threshold. My ephemeris places the 2027 transits 1.1–1.2 minutes before his predictions, within his 3.3–3.6-minute 1σ.

Star and search

L 198-93 = TIC 264151965 (Stassun et al. 2019; Paegert et al. 2021) has T = 10.95, Teff = 3430 K, R★ = 0.310 R ⊙ and M★ = 0.289 M ⊙ . Gaia DR3 gives G = 12.13, parallax 47.09 mas and RUWE 1.18 (Gaia Collaboration et al. 2023).

I searched TESS (Ricker et al. 2015) light curves of three pools of TIC v8.2 dwarfs: Teff = 2800–4000 K within 100 pc and either inside 12.5° of the north ecliptic pole (1,211 stars) or below ecliptic latitude –78° with logg > 4.0 (1,216), each with at least eight sectors; and Teff = 2700–4000 K within 25 pc outside the continuous viewing zones with at least three sectors (1,456, including this star). I dropped known TOI, CTOI and planet hosts, SPOC threshold-crossing-event (TCE) hosts and Eumemic (2026) candidates. Per sector I used SPOC 2-minute PDCSAP (Jenkins et al. 2016), else full-frame-image photometry from TESS-SPOC or QLP (Huang et al. 2020a, 2020b; Kunimoto et al. 2021); here, SPOC in Sectors 38 and 65, QLP in 99–102. Each was flattened with a 0.75 d wotan biweight filter (Hippke et al. 2019) and binned to 30 minutes. I searched 0.5–60 d (1–60 d beyond 25 pc) in up to four passes, 1,133 stars with Transit Least Squares (Hippke & Heller 2019) and 2,750 with the cuvarbase GPU box search (Hoffman 2022). The cuvarbase threshold, 21.7, is the highest score on flux-inverted light curves of 20 pool stars; the search recovered six known planets in TOI-700, TOI-2094 and TOI-2095.

Peaks above threshold with at least three transits went to the flux-level tests of LEO-vetter v1.2.0 (Kunimoto et al. 2025), which passed four known planets and failed 23 known artifacts. Three signals passed: an eclipsing binary, the candidate TIC 38699429, and this star (score 27.6).

Transit fit

I fitted the 2,501 10-minute points within 0.3 d of transit with batman (Kreidberg 2015) and emcee (Foreman-Mackey et al. 2013), with a circular orbit, fixed period, quadratic limb darkening fixed at u_1 = 0.25, u_2 = 0.35, and uniform priors (no stellar-density prior). Per-point errors are 875 ppm, 1.4826 times the out-of-transit median absolute deviation. It gives Rp/R★ = 0.0342 (+0.0015/–0.0011), b = 0.35 (+0.31/–0.24), duration 1.16 (+0.05/–0.04) h and Rp = 1.16 (+0.05/–0.04) R⊕. The fitted density, 24 (+6/–11) g cm^{-3}, agrees at about 1σ with 13.7 g cm^{-3} from the TIC mass and radius; fixing a/R★ to it raises the best χ^2 by 0.20 and gives b = 0.63 and Rp = 1.18 R⊕. Refitting with P free gives P = 4.8157979 ± 0.0000050 d and T0 = 2459336.1785 ± 0.0015 BJD_TDB. Murray's frozen period, 4.8158055 ± 0.0000057 d, differs by 1.0σ, and his 2027 transit times (epochs 72–82) lie 1.1–1.2 minutes after mine, within his 3.3–3.6-minute 1σ. His Rp/R★, 0.0339 (+0.0024/–0.0015), matches mine. At 0.0369 AU the insolation is 8.8 S⊕.

Vetting

A box fit (in-transit mean against the median 0.04–0.29 d from mid-transit) gives 1,255 ± 66 ppm (19.0σ). Odd and even depths are 1,258 ± 93 and 1,266 ± 93 ppm (0.06σ). Phase 0.5 gives –98 ± 76 ppm, and the strongest dip at any other phase is 3.7σ. Sector depths give χ^2 = 4.0 for 5 degrees of freedom about their weighted mean; the 29 single-transit depths give 43.2 for 28 (p = 0.03), none negative. Raw (SAP) depths agree with the corrected flux in all four sectors tested.

Weaker peaks at 0.594 d (later passes) and at 0.19816 d and 0.52390 d (transit-masked periodogram) match two VSX variables (Watson et al. 2006): the contact binary ASASSN-V J142912.22-584458.1 (76.7″; half-period 0.198 d, three times 0.594 d) and the RR Lyrae Gaia DR3 5890936706111098752 (24.5″, 0.5239 d).

In difference images from TESScut cutouts (Brasseur et al. 2019) in all six sectors, with per-pixel noise from 120 random-phase stacks, I fitted the TESS PRF (Bell & Higgins 2022) at the target and at all 733 Gaia DR3 stars within 150″ with G < 20.5. The target's fitted flux loss matches the depth in every sector (z = –0.3 to 2.0). For each neighbour, $\Delta\chi^2$ is the χ^2 with its PRF forced to the flux loss it would need minus that of a free-amplitude target fit, summed over sectors. Of the 733, 702 would need an eclipse deeper than 100% and 31 give $\Delta\chi^2 > 25$ (the closest, G = 18.98 at 6.4″, needs 79%; $\Delta\chi^2$ = 32). A free-position fit lands 2.6–15″ from the target in four sectors, but 71″ and 169″ away in Sectors 65 and 99 (5 and 4 transits).

Murray (2026a) found the signal in QLP Sectors 99–102, put the difference-image source 2.2 ± 1.7″ from the star and found FPP = 0.08. Eumemic (2026) rejected it because their highest-quality difference-image centroid lay 106.5″ from the target. My fits disagree: forcing the signal onto the G = 11.74 star at 106.8″ needs a 36% eclipse and gives $\Delta\chi^2 = 2 \times 10^7$.

False-positive probability

I ran TRICERATOPS v1.1.0 (Giacalone & Dressing 2020; Giacalone et al. 2021) ten times with 10^6 draws, SPOC apertures in Sectors 38 and 65, 5 × 5 pixel apertures elsewhere, all 2,129 field stars and no contrast curve. With no neighbours removed, FPP = 0.079 ± 0.004 and NFPP = 7.3×10^{-5}; 0.078 of

the FPP is a planet orbiting an unresolved bound companion. Companion limits came from MOLUSC (Wood et al. 2021), using its Gaia RUWE and contrast tests. Its 92,534 surviving companions (of 200,000) replaced TRICERATOPS's default prior for the bound-companion scenarios. With no neighbours removed, FPP = 0.020 ± 0.009 (0.011–0.033 across runs), still mostly that scenario. Both exceed the 0.015 validation threshold (Giacalone et al. 2021), and I have no high-resolution imaging, so I make no validation claim.

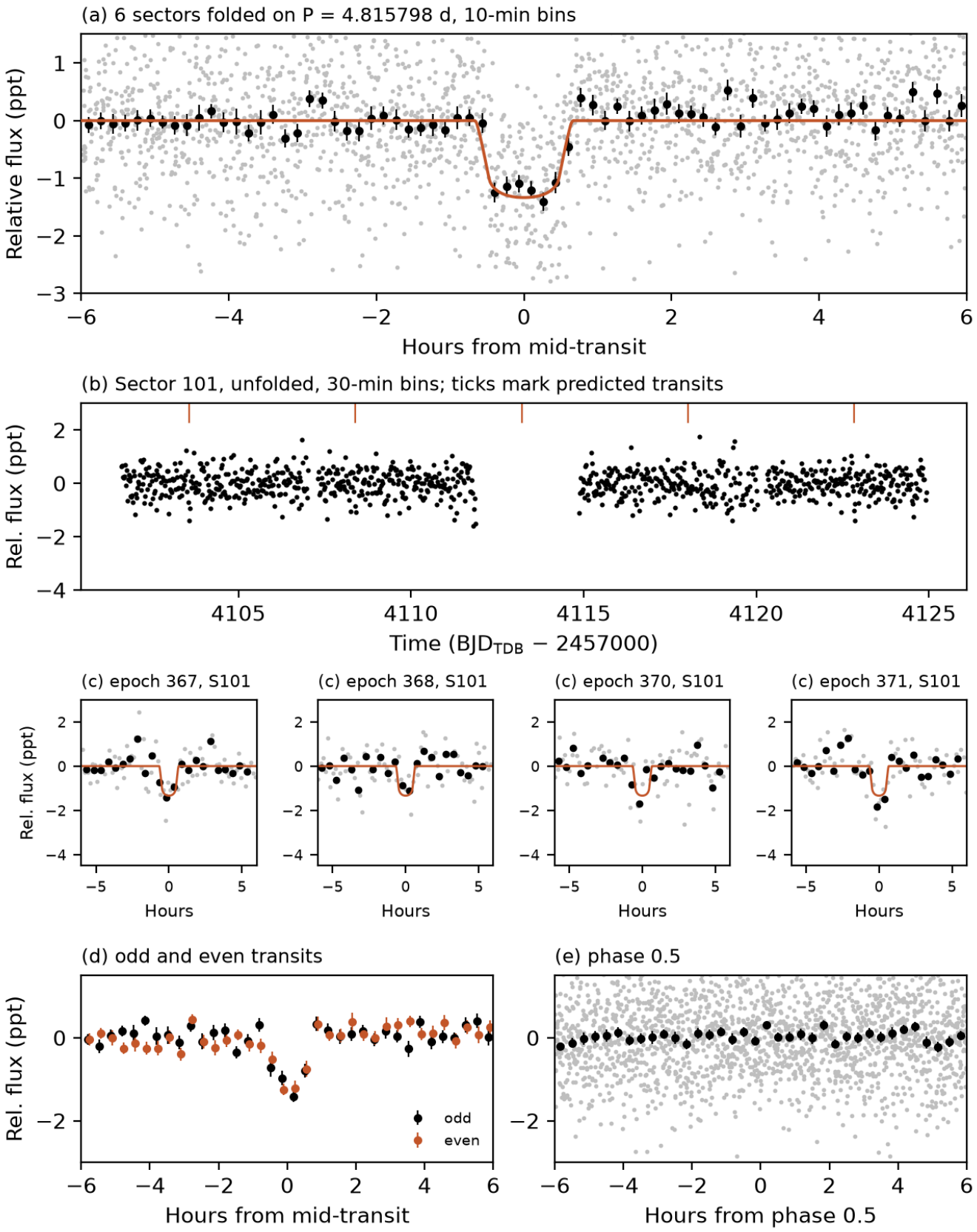
Summary

The L 198-93 candidate is a 1.2 $R_\oplus$ signal on a 4.8-day orbit around an M dwarf at 21 pc. Sectors 113–114 will test its ephemeris (Murray 2026a, 2026b). High-resolution imaging can test the bound-companion scenario, and ground-based photometry can place the transit on target.

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Figure 1



TESS photometry of L 198-93 (SPOC Sectors 38 and 65, QLP Sectors 99–102). (a) All six sectors folded on P = 4.815798 d (grey), 10-minute bins (black) and the median-parameter model (orange). (b) The unfolded, detrended Sector 101 light curve in 30-minute bins, with predicted transits marked; one falls in the data gap. (c) The four Sector 101 transits, 30-minute bins in black. (d) Odd (black) and even (orange) transits in 20-minute bins. (e) The light curve at phase 0.5.

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